

Work Note (v2) — Incorporating Published Literature Data into the Lab OCM Yield Model

To: Prof. Taniike **Topic:** A domain-adaptation study for C₂-yield prediction in Oxidative Coupling of Methane (OCM) **Supersedes:** version 1 of this note, which describes the problem setup, the two datasets and the five methods in full; that material is not repeated here. **Companion code:** `ocm_eval.py`, `taniike_validation.py`, `phase3_lit_prior.py`, `phase4_family_diagnosis.py`, `phase5_target_audit.py`, `phase6_our_experiments.py`, `phase6_candidates.py`, `phase7_prep_ood.py`, `phase8_target_robustness.py`, `phase9_equal_effort_eval.py`, `phase10_condition_subsampling.py`

What changed since version 1

Version 1 reported that a two-stage prior-feature method (“PFT”) improved C₂-yield prediction by 10.6 % over a lab-only baseline. **Following the stricter validation you proposed, that improvement does not survive, and we withdraw the claim.**

Claim in v1	Status in v2
PFT improves CV RMSE by 10.6 % (1.907 vs baseline 2.133)	Withdrawn. The gain was catalyst-identity leakage; under catalyst-grouped CV, PFT is 1.8 % worse than baseline
Literature data measurably helps the lab model	Not demonstrated in-domain. Four designs, all $\leq$ the composition-only control. It does help across preparation methods (§8)
Quantile normalisation is a necessary component	Not supported. Label treatment moves RMSE by 0.001–0.023, inside run-to-run noise at 3 seeds
—	New: the composition-only model screens <i>unseen</i> catalysts usefully ($\rho = 0.761$ over all 917 catalysts, 0.724 on the 771 with comparable measurement effort)
—	New: the Ba-family failure is mechanistically explained, and yields a data budget for new chemistry
—	New: one measurement per catalyst–temperature ranks nearly as well as all ~27 (§5)

The cause was specific: our Stage-1 expert was trained on literature data *together with the lab training rows*, so under a random row split it had seen the very catalysts it was later asked to help predict.

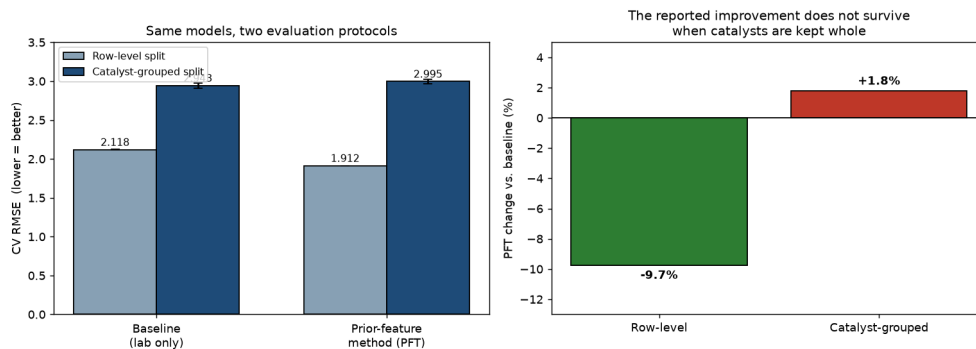


Figure 1 — The central finding. Identical models under two evaluation protocols. When every measurement of a catalyst is confined to one fold, the reported improvement inverts.

1. Evaluation methodology

Catalyst-grouped cross-validation is now our default. All measurements of a catalyst are assigned to exactly one fold, so every number answers: *how well do we predict a catalyst nobody has made yet?* Anything fitted on data — scaler, domain classifier, both model stages — sees training-fold data only.

Why the previous protocol was inadequate. The 89,074 measurements comprise only **917 distinct catalysts** at 5 temperatures, giving 4,399 (catalyst, temperature) cells of 20.2 rows on average. Your description of these rows as measurements *under different reaction conditions* is borne out by the row counts, and we should have taken it more literally. Cell sizes have a hard ceiling at exactly 27, a second at exactly $54 = 2 \times 27$, and nothing above; **15 catalysts hold exactly 135 rows, and all 15 decompose as exactly (27, 27, 27, 27, 27)** — that is **5 temperatures $\times$ 27 condition settings = 135**, matching the figure you gave us. We tested the competing reading that 27 was an export cut-off and rejected it: the spacing between the two lowest values in a 27-row cell is $2.03\times$ the interior spacing, whereas truncating a larger cell to 27 rows gives $0.90\times$ — a complete sample, not a truncated one.

These rows are therefore **not replicates**. Because the 27 settings are absent from the feature table, **19.9 % of total yield variance lies within cells** and cannot be reached from composition and temperature, flooring row-level RMSE at 1.757. That is a property of the missing columns, not a physical limit — recovering the condition data would make most of it learnable.

Version 1 reported 1.907, only 0.15 above that floor. In hindsight that should itself have prompted suspicion: a model cannot approach a floor that presumes knowledge of each catalyst's own cell means unless it has effectively memorised those catalysts. Point-wise RMSE is close to uninformative here, and it is precisely the metric a catalyst-identity leak flatters most. **Primary metrics are therefore catalyst-level, as you proposed:** Spearman correlation on maximum yield, and enrichment of true high performers among top-ranked predictions.

2. What the stricter validation showed

The improvement was leakage. Under catalyst-grouped CV: baseline **2.943**, PFT **2.995** (+1.8 %). The same two models under the row-level split give **2.118** and **1.912** (−9.7 %) — the difference between the two lines is leakage, not a change of model. Three readings of one 3-seed ablation grid point to the mechanism (not independent experiments): Stage 1 on literature *alone* reduces the row-level gain from −9.7 % to **−2.4 %** (QN prior) or **−2.7 %** (rank prior); under grouped CV the joint variant (**2.982**) is worse than literature-only (**2.938**); and literature-only is indistinguishable from baseline (**2.928**). (*The headline 2.995 is a 5-seed mean; 2.982 is the same configuration over the first 3 seeds.*)

Your quantile-normalisation hypothesis is supported, though not as strongly as we first stated. Gaps are 0.006 row-level, and catalyst-grouped 0.023 (QN vs raw) and 0.001 (QN vs rank) — the largest sits under our primary protocol and is close to the run-to-run spread there (per-configuration SD 0.022 and 0.035 over 3 seeds). The honest statement is that **no label treatment is distinguishable from another at this seed count**. The normalisation step can be dropped without measurable penalty.

With the leak closed we retested literature integration properly — a literature-only rank prior, similarity features, a gated prior, and a catalyst-level direct merge. Criteria were fixed before running. **None improved on composition alone.**

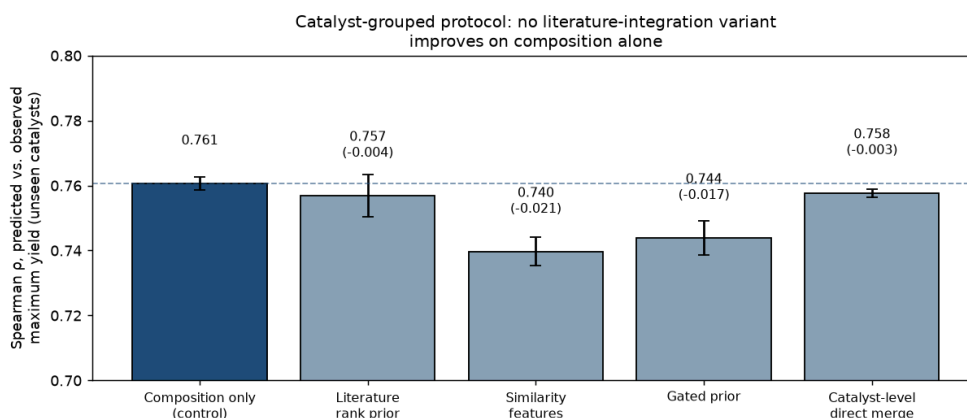


Figure 2 — No literature variant beats composition alone. Catalyst-grouped protocol; dashed line is the composition-only control.

We also tested a hypothesis of our own — that the prior might help where our coverage is thin, suggested by one family (Zr). Across **all 28 element families with ≥ 50 catalysts** the mean effect was **−0.0025**, 14/28 positive, strongest correlation with any coverage measure $|\rho| = 0.276$ against a pre-registered threshold of 0.5. The Zr result was selection from noise, and we discarded it.

3. Screening unseen catalysts

Rebuilt at catalyst level — composition → maximum yield, 917 training examples, no temperature — the model matches the full 89,074-row model on ranking while training on $\sim 100\times$ fewer rows.

Metric	All 917 catalysts	95 % CI	Equal-effort set (771)
Spearman ρ (predicted vs. observed max yield)	0.761	0.725 – 0.785	0.724
Enrichment of true top-decile among top-decile predicted	4.28×	3.04 – 4.89×	3.77×
Precision@20	0.44	0.15 – 0.65	0.35

With 92 catalysts in the top decile of 917, these are less precise than a point estimate suggests.

Why the second column exists, and why we consider it the honest one. Grid coverage in your data is coupled to performance — cells run further contain better yields — so a score over all 917 catalysts is partly a record of which experiments were completed. The *equal-effort set* is the 771 catalysts with ≥ 20 measurements in at least one cell; there the coupling is gone by measurement, with Spearman(measurement count, observed maximum) falling from **+0.293** to **+0.003**. Moving to it costs 0.037 Spearman and 0.51× enrichment.

That drop is not an artifact of scoring fewer catalysts: 300 **random** 771-catalyst subsets of the *same* predictions give 0.767 with a 95 % range of 0.756 – 0.780, and 0.724 lies below it.

A negative control worth more than either number. We refitted the identical model with the *number of measurements* as its target — it never sees a yield. That ranking reaches **Spearman 0.400** against observed maximum yield, but enrichment **0.87×**, no better than chance. Rank correlation is partly purchasable from experimental effort; enrichment is not. That is why we treat enrichment as primary.

One consequence for campaign design. Inside the model’s own top-ranked region — the only regime a campaign occupies — internal ordering carries little information: $\rho =$ **0.179** within the top 150 and **−0.066** within the top 20. The model *selects* well (its top 20 average 17.3 % observed maximum against ~10.5 % library-wide) but does not *order* within its selection. A shortlist is a set to test, not a league table.

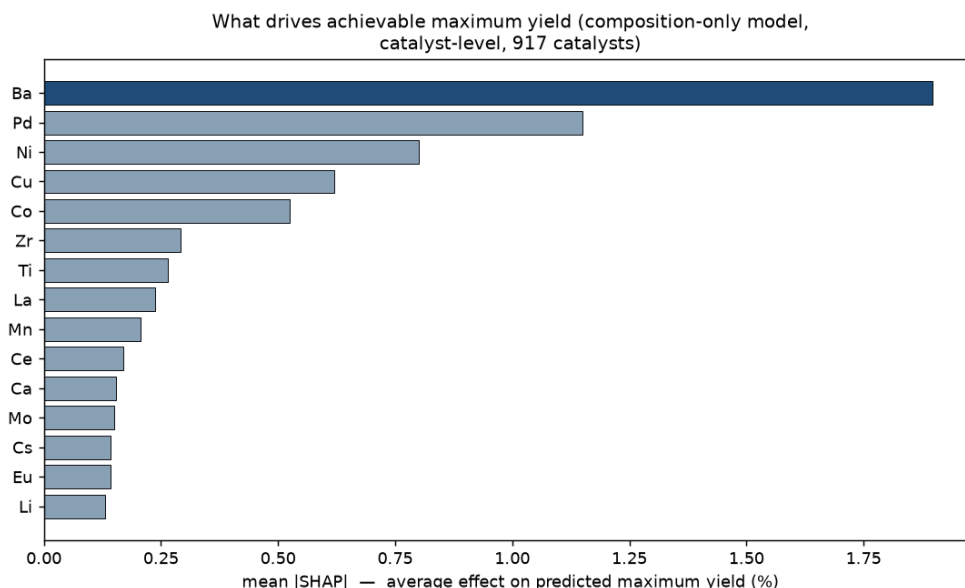


Figure 3 — What drives achievable maximum yield. Composition-only model. (Regenerated: the corresponding v1 figure ranked the literature prior first, because that model was the leaked pipeline.)

4. How much of the condition grid does a catalyst need?

Keep only k of the ~ 27 measurements per catalyst–temperature cell (drawn at random, 5 independent draws per k , always scored against the true maximum from the full data):

Measurements kept	Share of full grid	Spearman	Enrichment
1 per cell (~ 5 runs/catalyst)	5 %	0.759 ± 0.002	$3.99\times$
2 per cell (~ 10 runs/catalyst)	10 %	0.760 ± 0.002	$4.11\times$
3 per cell (~ 15 runs/catalyst)	15 %	0.765 ± 0.002	$4.24\times$
all ~ 27 per cell	100 %	0.761	$4.28\times$

One measurement per cell ranks almost as well as all 27, stable across five independent draws. Labels are biased low (-2.8 yield points at $k = 1$, below 0.4 by $k = 13$), but the bias is roughly uniform across catalysts, so ranking survives even though absolute yield estimates would not.

Temperature is different. 700°C alone gives $\rho = 0.346$, enrichment $1.07\times$ — no better than chance — because 70 % of catalysts reach their maximum at 800°C or above.

Redundancy is within a temperature, not across temperatures. Mechanism: between-catalyst spread in true maximum yield (SD 5.32) is $3.8\times$ the typical spread within one cell (SD 1.41).

This depends on the 27 slots being individually selectable. If they are successive samples from one continuous run, “keep 1 of 27” means stopping a run early, not choosing a condition, and the result would not transfer to a prospective design. See question 2 in §8.

5. Why Ba fails, and how much data a new family needs

Family holdouts behave reasonably for La, Ti, Zr and Ce (ρ 0.62–0.68) but poorly for Ba (0.526). Ba catalysts average **13.76 %** maximum yield against **8.95 %** for the rest, and **78 % of the lab's top decile contains Ba** — removing them removes the high-yield regime.

The mechanism is provable. With no Ba in training the Ba column is constant, so no tree splits on it, and retraining with that column **deleted entirely** gives **bit-identical predictions**. The model prices Ba catalysts as though Ba were absent, underpredicting the best by **9.8 yield points**. Ten random pseudo-families of equal size score 0.752, so this is label coverage, not sample size.

Family	ρ , none seen	ρ , fully seen	% of ceiling at zero
Ba	0.509	0.683	74.4 %
La	0.678	0.731	92.7 %
Ti	0.618	0.664	93.0 %
Zr	0.646	0.724	89.2 %
Ce	0.643	0.722	89.1 %

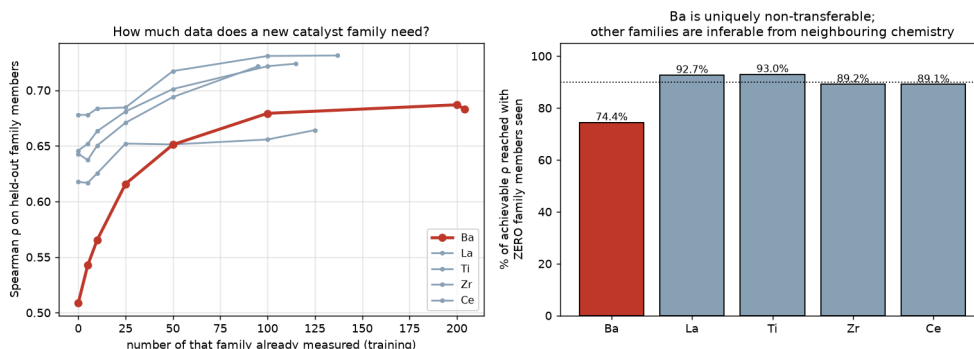


Figure 4 — A data budget for new chemistry. Left: performance against family members already measured. Right: fraction of achievable performance reached having seen none.

Ba is the most *consequential* family to lose, not the hardest to predict — five of the 28 score below it (Pd 0.217, Cu 0.286, Al 0.329, Ni 0.329, Co 0.493). What sets Ba apart is weight: it holds 78 % of the top decile and gains far more than any other family from seeing its own members (+0.175, against +0.047 to +0.079 elsewhere).

On the data budget, the threshold depends on the definition, and the two readings differ by five times. The Ba curve runs 0.509 (none seen) $\rightarrow$ 0.565 (10) $\rightarrow$ 0.616 (25) $\rightarrow$ 0.651 (50) $\rightarrow$ 0.683 (all 204). Taking 80 % of the **final level** (0.547) gives **10 members**; taking 80 % of the **gain** (0.648) gives roughly **50**. We now think the second is more meaningful: Ba already reaches 0.509 having seen no Ba at all, so the level-based bar is nearly cleared by seeing nothing. The same caution applies to the “0 needed” for La, Ti, Zr and Ce. Reaching 95 % takes about 50 for Ba and 25–50 for the others — order-of-magnitude guidance only, since for Ti and Zr the per-point seed spread (0.03–0.12) is comparable to the whole gain.

6. A candidate list for prospective validation

We enumerated **26,414 unseen candidates** in your own design grammar — impregnation, one support at ~90 % with 2–3 promoters at ~3.33 %, drawn from supports and promoters already in use — and scored them with a 10-seed ensemble. No literature prior is used. Every candidate carries a coverage flag, verified rather than assumed: for an element absent from our data (Ag), predictions with and without its column differ by exactly zero, confirming such candidates are unpriceable and must be flagged.

The highest-ranked candidate is **Ba(90) + Mo(3.33) + Zn(3.33) + Fe(3.33)**, predicted 18.79 %. We deliberately attach no error bar: the ± 0.09 our ensemble reports is only seed spread, while the model’s catalyst-level error on held-out catalysts is about **2.7 yield points** (MAE). The number ranks candidates; it does not forecast a yield. Absolute predictions also compress at the extreme — our maximum is 18.79 % against observed training yields reaching 21.50 %.

The top 20 are chemically monotonous (all Ba, mostly Mo), and a diversity constraint did not change this, because the model’s Ba preference is genuine. Best candidate per support:

Support	Ba	Ti	La	Ca	Mg	Si	Al	Zr	Ce
Best predicted max yield (%)	18.79	16.49	15.82	15.54	14.66	13.83	13.63	13.02	12.93

Our suggested campaign (17 catalysts, `campaign_shortlist.csv`) splits the budget: **Tier A**, 12 catalysts at the model’s optimum (18.33–18.79 %), where roughly **2–8 of the 12** would be genuine top-decile performers on the retrospective precision@20 CI of 0.15–0.65; and **Tier B**, 5 catalysts one per alternative support (Ti 16.49, La 15.82, Ca 15.54, Mg 14.66, Si 13.83). Tier B costs predicted yield and we do not expect it to win — it is an information purchase, since 78 % of the existing top decile already contains Ba, so the model’s preference may partly reflect coverage rather than chemistry. We defer to your judgement on synthesis feasibility.

A limit we should state plainly before you spend reactor time. Seventeen catalysts from the top-ranked region with no control arm cannot confirm or refute the model — per \$3, internal ordering there is close to uninformative. The list is a reasonable set to *try*; it is not a test.

If a test is wanted, a different allocation of the same reactor budget: measuring 5 conditions at each of 750, 800, 850 and 900 °C — 20 runs rather than 135 — reproduces the ranking of your 811 fully-measured catalysts at $\rho = 0.955$ (low by 1.31 yield points, a bias that can be pre-declared; §4 has the stability-checked version). That buys roughly **72 catalysts screened instead of 17** measured exhaustively, the best few then confirmed at full coverage, with part of the batch drawn at random as a control arm. The cost is more syntheses for the same reactor hours.

7. Relation to prior work

The mechanism of PFT — a model’s prediction used as an input feature — is **stacked generalisation** (Wolpert, *Neural Networks* 5, 1992, 241–259) applied across distributions. “Prior Feature Transfer” was our internal shorthand, not a standard term.

Our corrected understanding of when it can help. Because the prior $p = f(x)$ is a deterministic function of features the final model already has, $I(y; x, f(x)) = I(y; x)$ — it cannot add information. It can only supply an inductive bias, or exploit source data covering regions the target does not. Neither applies here at scale: ~79.7 % of the literature is out-of-distribution relative to the lab; quantile normalisation aligns marginal but not conditional distributions; and literature yields come from each paper’s own conditions, whereas our target is a maximum over a standardised battery. **The honest scope statement is that this family of methods is a local-coverage tool, not a global-information tool** — which is exactly what §8 then confirms.

Related approaches remain distinct: Δ -machine-learning (Ramakrishnan et al., *JCTC* 11, 2015, 2087) adds a source estimate as an additive baseline; importance weighting (Huang et al., NIPS 2006; Sugiyama et al., *JMLR* 8, 2007) keeps source labels in the objective — our DRST and KMM baselines are instances; label-shift correction (Lipton et al., ICML 2018) reweights the label distribution.

8. Limitations, open questions, and what we are doing next

- **The literature contribution is not demonstrated in-domain.** Four designs plus a 28-family follow-up all returned null. We report this rather than keep searching for a variant that scores.
- **Cross-preparation transfer is where literature data finally helps.** Predicting *impregnation* literature gives $p = 0.398$; *non-impregnation* gives 0.238, where the model cannot select at all (top-decile picks average 11.50 % against a population mean of 10.34 %, while the true top decile averages 21.92 % — enrichment 0.42 $\times$, worse than random). **Adding impregnation literature to training raises p from 0.238 to 0.388 (+0.150, 5/5 seeds).** Two caveats: absolute performance remains poor, and plain merging outperforms the prior-feature construction (0.388 vs 0.318) — the value is the data, not the two-stage machinery.
- **Novel promoter families cannot be priced.** Structural, not a modelling deficiency; §5 quantifies the data needed to remove it.
- **The target carries a measurement-effort confound.** 47 catalysts have fewer than 20 measurements and systematically low maxima. More generally, how much of the grid was run tracks how well the catalyst performed: Spearman(cell size, cell maximum yield) = **+0.441**, mean cell yield rising from 2.22 % in cells of 1–5 rows to 6.05 % in cells of 27. Only 811 of 917 catalysts have all five temperatures, and 186 cells are absent entirely. Excluding low-count catalysts changes our headline by 0.002, so nothing here hinges on it.

Three questions, in order of value to us:

1. **Do the ~27 condition settings per catalyst–temperature exist in retrievable form?** This is the largest single opportunity: the 19.9 % of variance now unreachable becomes largely learnable, condition-level modelling becomes meaningful, and row-level RMSE becomes a well-posed target again.

2. **Are those ~27 measurements distinct reaction conditions, or successive time-on-stream samples at a single condition?** We could not settle this from the file. If the latter, a catalyst's maximum is a fresh-catalyst transient rather than an achievable optimum — which changes both what our target means and whether §4's result transfers to a prospective design.
3. **Were incomplete runs stopped deliberately when results looked poor?** Coverage correlates with performance, and the answer decides whether that bias is correctable or is itself informative.

What we are doing meanwhile. Turning §4's subsampling result into a pre-registered screening protocol, and preparing the wider screen-then-confirm design in §6 — both pending your answer to question 2. We consider further in-domain literature-integration variants closed and are not pursuing more. Any prospective campaign would have its expected hit rate (precision@20, CI 0.15–0.65) pre-registered before results arrive.

All numbers here are stored outputs of the scripts named at the head of this note; each figure is generated from the corresponding experiment's JSON, so figures cannot drift from the experiments that produced them.